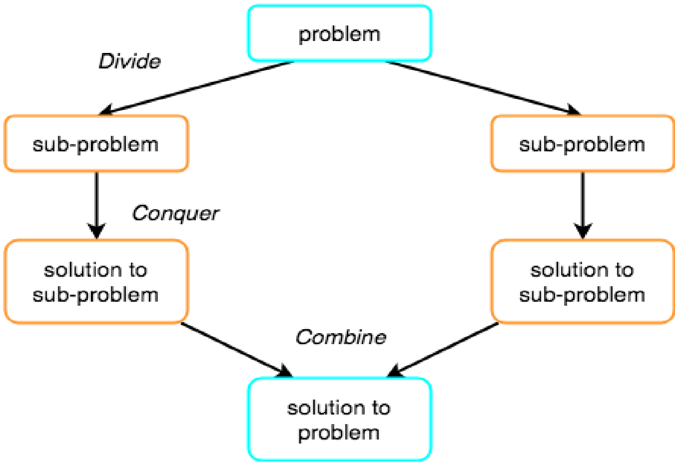
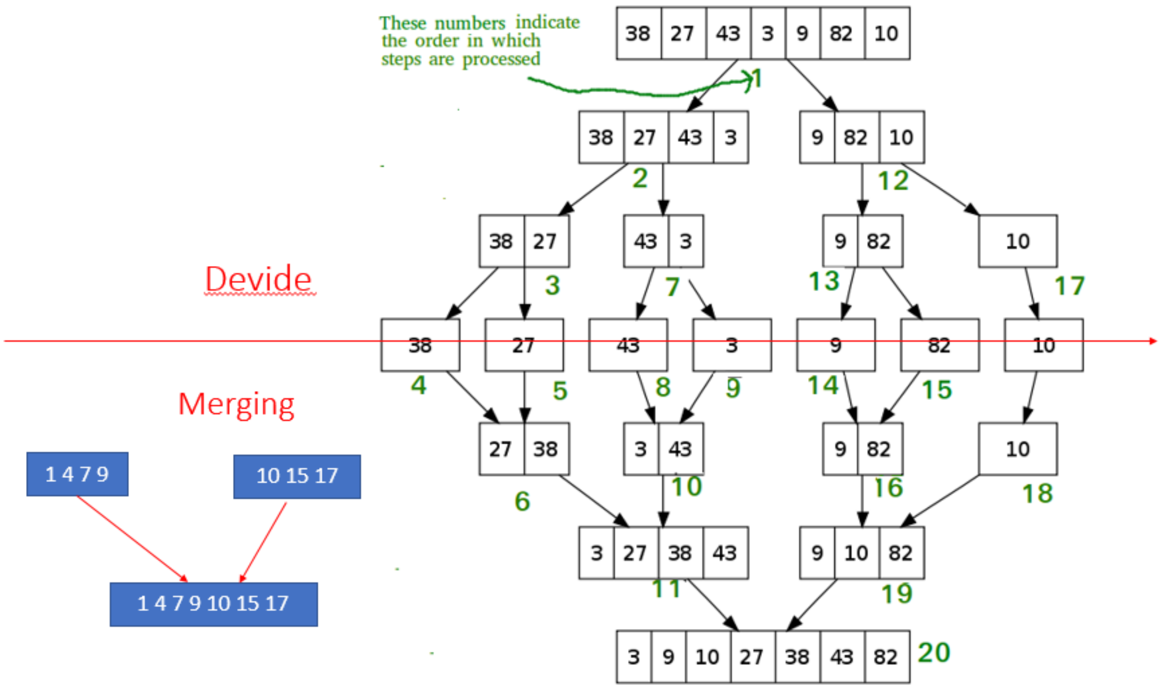
Merge sort

Merge sort works on the principle of Divide and Conquer Algorithmic approach. It involves three steps. Divide the problem into multiple small problems. Conquer the sub problems by solving them. The idea is to break down the problem into atomic sub problems, where they are actually solved. Combine the solutions of the sub problems to find the solution of the actual problem.



D&C Illustration

Merge sort Illustration:



Algorithm:

Merge(A, BEG, MID, END*)* *// **A[N]*** is an array of size N.

Step 1: *[INITIALIZE]SET I:= BEG, J:= MID+1, INDEX:=0*

Step 2: Repeat WHILE (I <= MID) AND (J <=END)

IF A[I] < A[J] THEN

SET OA[INDEX] := A[I]

SET I:= I +1

ELSE

SET OA[INDEX] := A[J]

SET J:= J +1

END IF

SET INDEX := INDEX + 1

[End of while]

Step 3:

IF I > MID THEN // Copy the remaining elements of right sub-array, if any

Repeat WHILE J <= END

SET OA[INDEX] := A[J]

SET INDEX := INDEX + 1.

SET J := J + 1.

[END OF WHILE]

ELSE // Copy the remaining elements of left sub-array, if any

Repeat WHILE I <= MID

SET OA[INDEX] := A[I]

SET INDEX := INDEX + 1.

SET I := I + 1.

[END OF WHILE]

[END OF IF]

Step 4: [Copy the contents of OA elements back to A] SET K:=0

Step 5: Repeat WHILE K < INDEX

SET A[K] := OA[K]

SET K := K+1

[END OF LOOP]

Step 6: End

MERGE_SORT(A, BEG, END)

Step 1: IF BEG < END THEN

SET MID := (BEG + END)/2.

CALL MERGE_SORT(A,BEG,MID)

CALL MERGE_SORT(A,MID+1, END)

MERGE(A, BEG, MID, END)

[END IF]

Step 2: exit